



Multiplication with a Reasonableness Check Teaching Bundle

Grade 5 • North Carolina • Mathematics

NC.5.NBT.5 • Use multi-digit multiplication accurately.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Use multi-digit multiplication accurately. This bundle practices a focused part of NC.5.NBT.5; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

Multiply by the ones and tens separately. Shift the tens product by one place. Add and compare with an estimate.

Example 1: The community center packs 107 cards in each of 22 boxes. Find the total. Show partial products and explain why they can be added.

$107 \times 20 + 107 \times 2 = 2140 + 214 = 2354$. The parts account for every box or card exactly once.

Example 2: The science club packs 148 cards in each of 12 boxes. Find the total. Show partial products and explain why they can be added.

$148 \times 10 + 148 \times 2 = 1480 + 296 = 1776$. The parts account for every box or card exactly once.

Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1-4

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

The science club packs 15 cards in each of 12 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 02

The science club packs 63 cards in each of 29 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 03

The music club packs 363 cards in each of 26 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 04

The music club packs 269 cards in each of 15 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

Task cards 5–8

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

The science club packs 32 cards in each of 27 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 06

The community center packs 67 cards in each of 17 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 07

The reading group packs 136 cards in each of 26 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 08

The design team packs 389 cards in each of 21 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.



Task cards 9–12

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

The reading group packs 35 cards in each of 28 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 10

The design team packs 61 cards in each of 18 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 11

The art club packs 199 cards in each of 23 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 12

The science club packs 295 cards in each of 17 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.



Task cards 13–16

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

The library team packs 44 cards in each of 19 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 14

The school garden packs 91 cards in each of 15 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 15

The music club packs 342 cards in each of 15 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.

CARD 16

The science club packs 182 cards in each of 19 boxes. Find the total. Show partial products and explain why they can be added.

Explain your method.



Task-card response record

Use this page for final answers; attach drawings or full worked solutions on separate paper.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____

Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. The art club packs 390 cards in each of 12 boxes. Find the total. Show partial products and explain why they can be added.

2. The library team packs 206 cards in each of 14 boxes. Find the total. Show partial products and explain why they can be added.



Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. The library team packs 12 cards in each of 17 boxes. Find the total. Show partial products and explain why they can be added.

4. The community center packs 77 cards in each of 26 boxes. Find the total. Show partial products and explain why they can be added.



Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. The science club packs 341 cards in each of 18 boxes. Find the total. Show partial products and explain why they can be added.

6. The music club packs 270 cards in each of 24 boxes. Find the total. Show partial products and explain why they can be added.



Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. The community center packs 48 cards in each of 16 boxes. Find the total. Show partial products and explain why they can be added.

8. The science club packs 86 cards in each of 23 boxes. Find the total. Show partial products and explain why they can be added.



Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. The science club packs 283 cards in each of 12 boxes. Find the total. Show partial products and explain why they can be added.

10. The music club packs 215 cards in each of 14 boxes. Find the total. Show partial products and explain why they can be added.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The school garden packs 258 cards in each of 23 boxes. Find the total. Show partial products and explain why they can be added.

2. The library team packs 208 cards in each of 25 boxes. Find the total. Show partial products and explain why they can be added.

3. The reading group packs 57 cards in each of 15 boxes. Find the total. Show partial products and explain why they can be added.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. The art club packs 31 cards in each of 16 boxes. Find the total. Show partial products and explain why they can be added.

5. The library team packs 330 cards in each of 23 boxes. Find the total. Show partial products and explain why they can be added.

6. The design team packs 219 cards in each of 19 boxes. Find the total. Show partial products and explain why they can be added.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. The library team packs 329 cards in each of 19 boxes. Find the total. Show partial products and explain why they can be added.

2. The school garden packs 347 cards in each of 26 boxes. Find the total. Show partial products and explain why they can be added.

3. The school garden packs 64 cards in each of 12 boxes. Find the total. Show partial products and explain why they can be added.

4. The library team packs 50 cards in each of 27 boxes. Find the total. Show partial products and explain why they can be added.



Task-card worked solutions

1. 180

$15 \times 10 + 15 \times 2 = 150 + 30 = 180$. The parts account for every box or card exactly once.

2. 1827

$63 \times 20 + 63 \times 9 = 1260 + 567 = 1827$. The parts account for every box or card exactly once.

3. 9438

$363 \times 20 + 363 \times 6 = 7260 + 2178 = 9438$. The parts account for every box or card exactly once.

4. 4035

$269 \times 10 + 269 \times 5 = 2690 + 1345 = 4035$. The parts account for every box or card exactly once.

5. 864

$32 \times 20 + 32 \times 7 = 640 + 224 = 864$. The parts account for every box or card exactly once.

6. 1139

$67 \times 10 + 67 \times 7 = 670 + 469 = 1139$. The parts account for every box or card exactly once.

Task-card worked solutions

7. 3536

$136 \times 20 + 136 \times 6 = 2720 + 816 = 3536$. The parts account for every box or card exactly once.

8. 8169

$389 \times 20 + 389 \times 1 = 7780 + 389 = 8169$. The parts account for every box or card exactly once.

9. 980

$35 \times 20 + 35 \times 8 = 700 + 280 = 980$. The parts account for every box or card exactly once.

10. 1098

$61 \times 10 + 61 \times 8 = 610 + 488 = 1098$. The parts account for every box or card exactly once.

11. 4577

$199 \times 20 + 199 \times 3 = 3980 + 597 = 4577$. The parts account for every box or card exactly once.

12. 5015

$295 \times 10 + 295 \times 7 = 2950 + 2065 = 5015$. The parts account for every box or card exactly once.

Task-card worked solutions

13. 836

$44 \times 10 + 44 \times 9 = 440 + 396 = 836$. The parts account for every box or card exactly once.

14. 1365

$91 \times 10 + 91 \times 5 = 910 + 455 = 1365$. The parts account for every box or card exactly once.

15. 5130

$342 \times 10 + 342 \times 5 = 3420 + 1710 = 5130$. The parts account for every box or card exactly once.

16. 3458

$182 \times 10 + 182 \times 9 = 1820 + 1638 = 3458$. The parts account for every box or card exactly once.

Warm-up worked solutions

1. 4680

$390 \times 10 + 390 \times 2 = 3900 + 780 = 4680$. The parts account for every box or card exactly once.

2. 2884

$206 \times 10 + 206 \times 4 = 2060 + 824 = 2884$. The parts account for every box or card exactly once.

3. 204

$12 \times 10 + 12 \times 7 = 120 + 84 = 204$. The parts account for every box or card exactly once.

4. 2002

$77 \times 20 + 77 \times 6 = 1540 + 462 = 2002$. The parts account for every box or card exactly once.

5. 6138

$341 \times 10 + 341 \times 8 = 3410 + 2728 = 6138$. The parts account for every box or card exactly once.

6. 6480

$270 \times 20 + 270 \times 4 = 5400 + 1080 = 6480$. The parts account for every box or card exactly once.

Warm-up worked solutions

7. 768

$48 \times 10 + 48 \times 6 = 480 + 288 = 768$. The parts account for every box or card exactly once.

8. 1978

$86 \times 20 + 86 \times 3 = 1720 + 258 = 1978$. The parts account for every box or card exactly once.

9. 3396

$283 \times 10 + 283 \times 2 = 2830 + 566 = 3396$. The parts account for every box or card exactly once.

10. 3010

$215 \times 10 + 215 \times 4 = 2150 + 860 = 3010$. The parts account for every box or card exactly once.

Classwork worked solutions

1. 5934

$258 \times 20 + 258 \times 3 = 5160 + 774 = 5934$. The parts account for every box or card exactly once.

2. 5200

$208 \times 20 + 208 \times 5 = 4160 + 1040 = 5200$. The parts account for every box or card exactly once.

3. 855

$57 \times 10 + 57 \times 5 = 570 + 285 = 855$. The parts account for every box or card exactly once.

4. 496

$31 \times 10 + 31 \times 6 = 310 + 186 = 496$. The parts account for every box or card exactly once.

5. 7590

$330 \times 20 + 330 \times 3 = 6600 + 990 = 7590$. The parts account for every box or card exactly once.

6. 4161

$219 \times 10 + 219 \times 9 = 2190 + 1971 = 4161$. The parts account for every box or card exactly once.

Exit-ticket worked solutions

1. 6251

$329 \times 10 + 329 \times 9 = 3290 + 2961 = 6251$. The parts account for every box or card exactly once.

2. 9022

$347 \times 20 + 347 \times 6 = 6940 + 2082 = 9022$. The parts account for every box or card exactly once.

3. 768

$64 \times 10 + 64 \times 2 = 640 + 128 = 768$. The parts account for every box or card exactly once.

4. 1350

$50 \times 20 + 50 \times 7 = 1000 + 350 = 1350$. The parts account for every box or card exactly once.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The October book drive packs 57 cards in each of 26 boxes. Find the total. Show partial products and explain why they can be added.

2. The September art display packs 95 cards in each of 28 boxes. Find the total. Show partial products and explain why they can be added.

3. The September art display packs 360 cards in each of 23 boxes. Find the total. Show partial products and explain why they can be added.

4. The fall garden club packs 150 cards in each of 25 boxes. Find the total. Show partial products and explain why they can be added.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The October book drive packs 67 cards in each of 15 boxes. Find the total. Show partial products and explain why they can be added.

6. The fall garden club packs 66 cards in each of 24 boxes. Find the total. Show partial products and explain why they can be added.

7. The autumn reading fair packs 149 cards in each of 22 boxes. Find the total. Show partial products and explain why they can be added.

8. The autumn reading fair packs 398 cards in each of 26 boxes. Find the total. Show partial products and explain why they can be added.



Fall • Worked solutions

1. 1482

$57 \times 20 + 57 \times 6 = 1140 + 342 = 1482$. The parts account for every box or card exactly once.

2. 2660

$95 \times 20 + 95 \times 8 = 1900 + 760 = 2660$. The parts account for every box or card exactly once.

3. 8280

$360 \times 20 + 360 \times 3 = 7200 + 1080 = 8280$. The parts account for every box or card exactly once.

4. 3750

$150 \times 20 + 150 \times 5 = 3000 + 750 = 3750$. The parts account for every box or card exactly once.

Fall • Worked solutions

5. 1005

$67 \times 10 + 67 \times 5 = 670 + 335 = 1005$. The parts account for every box or card exactly once.

6. 1584

$66 \times 20 + 66 \times 4 = 1320 + 264 = 1584$. The parts account for every box or card exactly once.

7. 3278

$149 \times 20 + 149 \times 2 = 2980 + 298 = 3278$. The parts account for every box or card exactly once.

8. 10348

$398 \times 20 + 398 \times 6 = 7960 + 2388 = 10348$. The parts account for every box or card exactly once.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The winter reading club packs 248 cards in each of 28 boxes. Find the total. Show partial products and explain why they can be added.

2. The winter supply drive packs 210 cards in each of 16 boxes. Find the total. Show partial products and explain why they can be added.

3. The indoor garden project packs 18 cards in each of 14 boxes. Find the total. Show partial products and explain why they can be added.

4. The January puzzle club packs 55 cards in each of 16 boxes. Find the total. Show partial products and explain why they can be added.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The December art display packs 137 cards in each of 13 boxes. Find the total. Show partial products and explain why they can be added.

6. The January puzzle club packs 299 cards in each of 14 boxes. Find the total. Show partial products and explain why they can be added.

7. The indoor garden project packs 62 cards in each of 14 boxes. Find the total. Show partial products and explain why they can be added.

8. The winter supply drive packs 74 cards in each of 17 boxes. Find the total. Show partial products and explain why they can be added.



Winter • Worked solutions

1. 6944

$248 \times 20 + 248 \times 8 = 4960 + 1984 = 6944$. The parts account for every box or card exactly once.

2. 3360

$210 \times 10 + 210 \times 6 = 2100 + 1260 = 3360$. The parts account for every box or card exactly once.

3. 252

$18 \times 10 + 18 \times 4 = 180 + 72 = 252$. The parts account for every box or card exactly once.

4. 880

$55 \times 10 + 55 \times 6 = 550 + 330 = 880$. The parts account for every box or card exactly once.

Winter • Worked solutions

5. 1781

$137 \times 10 + 137 \times 3 = 1370 + 411 = 1781$. The parts account for every box or card exactly once.

6. 4186

$299 \times 10 + 299 \times 4 = 2990 + 1196 = 4186$. The parts account for every box or card exactly once.

7. 868

$62 \times 10 + 62 \times 4 = 620 + 248 = 868$. The parts account for every box or card exactly once.

8. 1258

$74 \times 10 + 74 \times 7 = 740 + 518 = 1258$. The parts account for every box or card exactly once.

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The seedling club packs 51 cards in each of 28 boxes. Find the total. Show partial products and explain why they can be added.

2. The April science team packs 13 cards in each of 17 boxes. Find the total. Show partial products and explain why they can be added.

3. The seedling club packs 182 cards in each of 25 boxes. Find the total. Show partial products and explain why they can be added.

4. The spring reading fair packs 376 cards in each of 29 boxes. Find the total. Show partial products and explain why they can be added.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The March art display packs 62 cards in each of 22 boxes. Find the total. Show partial products and explain why they can be added.

6. The seedling club packs 40 cards in each of 19 boxes. Find the total. Show partial products and explain why they can be added.

7. The March art display packs 281 cards in each of 27 boxes. Find the total. Show partial products and explain why they can be added.

8. The seedling club packs 185 cards in each of 19 boxes. Find the total. Show partial products and explain why they can be added.



Spring break • Worked solutions

1. 1428

$51 \times 20 + 51 \times 8 = 1020 + 408 = 1428$. The parts account for every box or card exactly once.

2. 221

$13 \times 10 + 13 \times 7 = 130 + 91 = 221$. The parts account for every box or card exactly once.

3. 4550

$182 \times 20 + 182 \times 5 = 3640 + 910 = 4550$. The parts account for every box or card exactly once.

4. 10904

$376 \times 20 + 376 \times 9 = 7520 + 3384 = 10904$. The parts account for every box or card exactly once.



Spring break • Worked solutions

5. 1364

$62 \times 20 + 62 \times 2 = 1240 + 124 = 1364$. The parts account for every box or card exactly once.

6. 760

$40 \times 10 + 40 \times 9 = 400 + 360 = 760$. The parts account for every box or card exactly once.

7. 7587

$281 \times 20 + 281 \times 7 = 5620 + 1967 = 7587$. The parts account for every box or card exactly once.

8. 3515

$185 \times 10 + 185 \times 9 = 1850 + 1665 = 3515$. The parts account for every box or card exactly once.

Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. The June art workshop packs 190 cards in each of 29 boxes. Find the total. Show partial products and explain why they can be added.

2. The summer library club packs 277 cards in each of 12 boxes. Find the total. Show partial products and explain why they can be added.

3. The summer library club packs 52 cards in each of 23 boxes. Find the total. Show partial products and explain why they can be added.

4. The summer supply team packs 80 cards in each of 24 boxes. Find the total. Show partial products and explain why they can be added.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. The July puzzle camp packs 376 cards in each of 23 boxes. Find the total. Show partial products and explain why they can be added.

6. The summer supply team packs 333 cards in each of 22 boxes. Find the total. Show partial products and explain why they can be added.

7. The summer supply team packs 62 cards in each of 23 boxes. Find the total. Show partial products and explain why they can be added.

8. The community garden packs 42 cards in each of 29 boxes. Find the total. Show partial products and explain why they can be added.



Summer • Worked solutions

1. 5510

$190 \times 20 + 190 \times 9 = 3800 + 1710 = 5510$. The parts account for every box or card exactly once.

2. 3324

$277 \times 10 + 277 \times 2 = 2770 + 554 = 3324$. The parts account for every box or card exactly once.

3. 1196

$52 \times 20 + 52 \times 3 = 1040 + 156 = 1196$. The parts account for every box or card exactly once.

4. 1920

$80 \times 20 + 80 \times 4 = 1600 + 320 = 1920$. The parts account for every box or card exactly once.



Summer • Worked solutions

5. 8648

$376 \times 20 + 376 \times 3 = 7520 + 1128 = 8648$. The parts account for every box or card exactly once.

6. 7326

$333 \times 20 + 333 \times 2 = 6660 + 666 = 7326$. The parts account for every box or card exactly once.

7. 1426

$62 \times 20 + 62 \times 3 = 1240 + 186 = 1426$. The parts account for every box or card exactly once.

8. 1218

$42 \times 20 + 42 \times 9 = 840 + 378 = 1218$. The parts account for every box or card exactly once.

Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Use multi-digit multiplication accurately. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. Multiply by the ones and tens separately. Shift the tens product by one place. Add and compare with an estimate.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The classroom supply team packs 48 cards in each of 18 boxes. Find the total. Show partial products and explain why they can be added.

2. The student design group packs 54 cards in each of 15 boxes. Find the total. Show partial products and explain why they can be added.

3. The school art team packs 291 cards in each of 23 boxes. Find the total. Show partial products and explain why they can be added.

4. The library helpers packs 170 cards in each of 14 boxes. Find the total. Show partial products and explain why they can be added.



Substitute lesson • Worked solutions

1. 864

$48 \times 10 + 48 \times 8 = 480 + 384 = 864$. The parts account for every box or card exactly once.

2. 810

$54 \times 10 + 54 \times 5 = 540 + 270 = 810$. The parts account for every box or card exactly once.

3. 6693

$291 \times 20 + 291 \times 3 = 5820 + 873 = 6693$. The parts account for every box or card exactly once.

4. 2380

$170 \times 10 + 170 \times 4 = 1700 + 680 = 2380$. The parts account for every box or card exactly once.

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Use multi-digit multiplication accurately. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. Multiply by the ones and tens separately. Shift the tens product by one place. Add and compare with an estimate.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. The quiet design team packs 83 cards in each of 24 boxes. Find the total. Show partial products and explain why they can be added.

2. The quiet design team packs 27 cards in each of 24 boxes. Find the total. Show partial products and explain why they can be added.

3. The quiet design team packs 112 cards in each of 25 boxes. Find the total. Show partial products and explain why they can be added.

4. The quiet design team packs 359 cards in each of 17 boxes. Find the total. Show partial products and explain why they can be added.



Emergency substitute lesson • Worked solutions

1. 1992

$83 \times 20 + 83 \times 4 = 1660 + 332 = 1992$. The parts account for every box or card exactly once.

2. 648

$27 \times 20 + 27 \times 4 = 540 + 108 = 648$. The parts account for every box or card exactly once.

3. 2800

$112 \times 20 + 112 \times 5 = 2240 + 560 = 2800$. The parts account for every box or card exactly once.

4. 6103

$359 \times 10 + 359 \times 7 = 3590 + 2513 = 6103$. The parts account for every box or card exactly once.

A completed lesson-plan example

Level and focus

Grade 5; NC.5.NBT.5; Use multi-digit multiplication accurately.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.5.NBT.5 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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